PHARMACOEPIDEMIOLOGY AND DRUG SAFETY 2001; **1** : 483–486
DOI: 10.1002/pds.677

ORIGINAL REPORT

Use of proportional reporting ratios (PRRs) for signal generation from spontaneous adverse drug reaction reports

S. J. W. Evans, P. C. Waller* and S. Davis

Post-Licensing Division Medicines Control Agency Market Towers 1 Nine Elms Lane London SW8 5NQ

SUMMARY

Background The process of generating 'signals' of possible unrecognized hazards from spontaneous adverse drug reaction reporting data has been likened to looking for a needle in a haystack. However, statistical approaches to the data have been underutilised.

Methods Using the UK Yellow Card database, we have developed and evaluated a statistical aid to signal generation called a Proportional Reporting Ratio (PRR). The proportion of all reactions to a drug which are for a particular medical condition of interest is compared to the same proportion for all drugs in the database, in a 2 2 table. We investigated a group of newly-marketed drugs using as minimum criteria for a signal, 3 or more cases, PRR at least 2, chi-squared of at least 4.

Findings The database was used to examine retrospectively 15 drugs newly-marketed in the UK, with the highest levels of ADR reporting. The method identified 481 signals meeting the minimum criteria during the period 1996–8. Further evaluation of these showed that 70 were known adverse reactions, 13 were events which were likely to be related to the underlying disease and 17 were signals requiring further evaluation.

Implications Proportional reporting ratios are a valuable aid to signal generation from spontaneous reporting data which are easy to calculate and interpret, and various refinements are possible. Crown copyright 2001. Reproduced with the permission of Her Majesty's Stationery Office. Published by John Wiley & Sons, Ltd.

KEY WORDS — drug safety; pharmacovigilance; signal generation; spontaneous AD reporting

INTRODUCTION

The primary purpose of spontaneous adverse drug reaction (ADR) reporting is to provide early warnings of hazards which have not been recognized prior to marketing of a drug because of limitations of clinical trials in respect of sample size, duration and generalisability to ordinary practice. The process of scrutinising spontaneous ADR data for hazards is known as signal generation. Once a signal has been generated, other sources of data are investigated and, if there is sufficient evidence of a public health issue, steps are taken by regulatory authorities aimed at minimising the risks and informing users.[1]

In the UK, reports of suspected ADRs are made on Yellow Cards by health professionals and are sent to the Medicines Control Agency (MCA) where they are entered onto the Adverse Drug Reactions On-line Information Tracking (ADROIT) database.[2] This database, which came into operation in 1991, contains more than 350 000 reports received from the beginning of the Yellow Card scheme in 1964, describing about 600 000 reactions.

If an adverse reaction is a rare disease then a signal may be generated by a small series of around 3–5 reports. However, in general, there is no easy answer to the question 'How many reports constitute a signal?'[3] This involves judgements being made based on the number and quality of the case reports, the nature of the adverse reaction, type of drug and level

* Correspondence to: Dr P. C. Waller, Post-Licensing Division, Medicines Control Agency, Market Towers, 1, Nine Elms Lane, London SW8 5NQ. Fax: 0207-273-0205.

Received 2 November 2001
Revised 7 November 2001
Accepted 9 November 2001
Published online: 10 December 2001

Crown copyright 2001. Reproduced with the permission of HMSO

of usage. There is a need to develop tools to aid and improve this process, and in particular to ensure that recognition of important signals is not delayed because the needle cannot be seen in the haystack.[4]

In this paper we describe a statistical aid to signal generation, the value of which we have explored in the ADROIT database.

METHODS

In order to judge whether the number of cases reported spontaneously exceeds what might be expected through a combination of chance and background 'noise', there are two accepted approaches.[5] The first involves using denominator data related to use of the drug (often dispensed prescriptions or sales as a proxy) and calculating reporting rates (reporting rate = no. of reports/no. of prescriptions). Comparisons between drugs based on reporting rates may be biased by many factors such as increased reporting for new drugs, and the effects of calendar time and publicity. Little weight can be placed on small differences in reporting rates between drugs but large differences (i.e. several-fold) may represent a signal worthy of investigation. The second approach is to use the total number of reports for the drug as a denominator and to calculate the proportion of all reactions which are the type of interest (e.g. hepatitis). This proportion may be compared with the value for other drugs and it is also possible to compare complete profiles of ADR reporting for drugs, where differences in the profile may represent signals. This proportionate approach has some advantages over reporting rates. First, no external data are needed and the limitations of such data (including delay in receipt) do not apply. Secondly, it may be expected to counteract some of the biases related to variable reporting. For example, if the overall level of reporting is high because of new drug bias, this will not necessarily affect the proportion of all reactions for the drug which are of a specified type.

We have developed a statistical aid to signal generation based on the proportionate approach which also utilises the stability of a large database. This involves calculation of the proportions of specified reactions or groups of reactions for drugs of interest where the comparator is all other drugs in the database. The result of such a calculation is called a proportional reporting ratio (PRR) where the PRR is $a/(a+c)$ divided by $b/(b+d)$ in a two by two table (see Table 1).

An example of such a calculation is given in Table 2. The expected or null value for a PRR is one and the values generated are measures related to strength of association which behave in a similar fashion to relative risks (the higher the PRR, the greater the strength of the signal). It is also possible to measure the size of the association using an odds ratio. Measures of statistical association may be calculated using a chi-squared test on one degree of freedom with Yates's correction. Judgement about whether or not there is a signal, and its strength, is made on the basis of three pieces of information i.e. the PRR, value of chi-squared and the absolute number of reports. An equivalent alternative to chi-squared is to calculate a confidence interval around the PRR.

Table 1. Calculation of PRRs

	Drug of interest	All other drugs in database
Reaction(s) of interest	a	b
All other reactions	c	d

Table 2. Example of a PRR calculation-rifabutin and uveitis

	Rifabutin	All other drugs
Uveitis	41	754
All other ADRs	14	591 958
TOTAL	55	592 712

PRR = 41/55 divided by 754/592, 712 = 586.
Chi-squared (1 df) = 22 740.

We tested the criteria using accumulated data for 15 newly-marketed drugs with the highest levels of ADR reporting during the period 1996–8. We examined whether the method would identify known hazards and possible adverse reactions which had not been previously recognized. A signal was defined as a PRR of at least 2, chi-squared of at least 4 and three or more cases. Using the ADROIT medical dictionary,[6] we calculated the PRRs for all reported preferred terms and asked the assessor responsible for monitoring the drug to evaluate all the signals meeting the above criteria to decide whether the ADR was recognized (on the basis of authorised product information), likely to be related to the underlying disease or a new signal worthy of investigation.

All calculations were performed using a standard statistical software program (STATA, Stata Corp, College Station, Texas).

RESULTS

In total, the method identified 487 potential signals meeting the minimum criteria (i.e. 10 of reported

Table 3. Evaluation of potential signals meeting minimum criteria for 15 newly-marketed drugs during the period 1996–8

Rank	Drug substance	Number of preferred terms reported	Number of potential signals* (of number of PTs reported)	R	E	S
1	Venlafaxine	532	64 (12)	52	10	2
2	Tramadol	390	38 (10)	25	0	13
3	Lamotrigine	505	47 (10)	24	8	15
4	Losartan	290	26 (9)	14	4	8
5	Nefazodone	356	50 (14)	37	7	6
6	Meloxicam	317	41 (13)	41	0	0
7	Alendronic acid	300	37 (12)	32	2	3
8	Dexfenfluramine	250	31 (12)	20	0	11
9	Moclobemide	269	26 (11)	7	15	4
10	Atorvastatin	264	28 (11)	27	0	1
11	Citalopram	278	24 (9)	16	1	7
12	Olanzapine	306	46 (15)	27	10	9
13	Montelukast	255	15 (6)	10	4	1
14	Mirtazapine	258	6 (2)	6	0	0
15	Nicorandil	212	2 (1)	1	1	0
Totals			481 (10)	339 (70)	62 (13)	80 (17)

The table excludes 6 unevaluable signals; *3 or more cases, PRR 2 or more, chi square 4 or more.
R = recognized adverse reaction, E = event considered to be related to underlying disease, S = signal requiring further evaluation.

preferred terms), of which 6 were excluded as unevaluable. The 15 drugs evaluated retrospectively, the total numbers of preferred terms reported for each drug, the numbers of signals identified and results of the evaluation are shown in Table 3. Of the 481 evaluable potential signals, 339 (70) were recognized adverse reactions, 62 (13) were considered to be events related to the underlying disease and 80 (17) were signals requiring further evaluation. On average, the method identified about 5 unrecognised signals per drug (each of which had a high level of reporting). To date, of the 80 new signals identified, 22 (28) have warranted detailed review, in 3 cases leading to a request for the manufacturer to change the product information. Of the remainder, 22 are being kept under review and no further action is proposed in relation to 36. The latter group includes 11 signals for dexfenfluramine, which was withdrawn from the market during the study period. The 339 recognized adverse reactions identified represented 71 of all the adverse reactions for these drugs listed in the 'undesirable effects' section of the Summary of Product Characteristics.

DISCUSSION

The mathematical basis of Proportional Reporting Ratios (PRRs) is straightforward and has been applied in other contexts where there are difficulties with denominators (e.g. proportional mortality ratios). A similar approach was developed by Finney in 1974[7], but has not been in general use subsequently.

Using data relating to recently marketed drugs, we found that PRRs within the ADROIT database for well-recognized ADRs were often strikingly high (e.g. more than 500 for uveitis associated with rifabutin, Table 2). Such problems are unlikely to be missed or even be detected much earlier using this approach. The main value is likely to be in highlighting a small series of cases where a PRR value of say 3–5 suggests a need for detailed evaluation of the series and further investigation. Conversely a PRR value close to or less than one may prevent unnecessary effort in evaluating case series which are the consequence of background 'noise' and do not truly represent a signal. The use of PRRs on a prospective basis is now performed routinely in the MCA.

Many potential signals are likely to be present in a large ADR database such as ADROIT. It is clear from our study that systematic use of PRRs identifies a high proportion of all recognised ADRs and also some new signals which might go unrecognised, at least until further cases are reported. The approach is not a substitute for detailed review of cases but an aid to deciding which series of cases most warrant further review. PRRs and chi-squared values are measures of association and not causality, and it is not surprising that some of signals identified were considered to be events unrelated to treatment.

Crown copyright 2001. Reproduced with the permission of HMSO